Unit 3 · Outcome 1 · Maintaining Biodiversity

The three categories of genes, species & ecosystems

KK01 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

The three categories of genes, species & ecosystems

U3·O1 · KK01 · Maintaining Biodiversity The three categories of genes , species & ecosystems Biodiversity isn't one thing you count — it's the variety of all life , measured at three nested levels. Get these three straight and the whole Biodiversity unit clicks into place. Drag the model below to explore how genes sit inside species, and species inside ecosystems. · Genetic — within a species Species — across species Ecosystem — across habitats 3D here, but every idea below works without it. Genetic diversity sits inside species diversity, which sits ins

What is biodiversity?

02 — definition What is biodiversity? Biodiversity (biological diversity) is the variety of all living things on Earth — every organism, the genes they carry, and the ecosystems they form. The trap is to read "variety" as "how many animals." Biodiversity is richer than a head-count: scientists describe it at three interconnected levels . Genetic Variety of genes (alleles) within a species or population. scope: inside one species Species Variety of different species living in an area. scope: across species Ecosystem Variety of ecosystems / habitats

Genetic diversity

03 — level one Genetic diversity Definition The variety of genes (alleles) within a species or a population — the differences in heritable traits between individuals of the same species. Think of all the eastern grey kangaroos in one mob: they share a species, but differ slightly in size, colour, immune-system genes and disease resistance. That spread of alleles is the population's genetic diversity. Why it matters It's the raw material for evolution . Natural selection can only act on variation that already exists. A genetically diverse population is mo

Species diversity

04 — level two Species diversity Definition The variety of different species in an area. It has two components — don't stop at the first one. # Species richness The number of different species present. "how many kinds?" Species evenness How evenly individuals are spread across those species (relative abundance). "how balanced?" Species diversity = richness + evenness. Two surveys can record the same number of species, yet differ in diversity because one is dominated by a single species. We'll prove this in the Richness vs evenness model — and it's the

Ecosystem diversity

05 — level three Ecosystem diversity Definition The variety of ecosystems across a region — the range of different habitats, communities and the ecological processes that happen within them. An ecosystem is a community of organisms plus their physical environment, interacting as a unit. Ecosystem diversity zooms right out: how many different ecosystems does a landscape contain? High ecosystem diversity: Victoria alone spans alpine ash forests, mallee, temperate wetlands, native grasslands and rocky coasts — each a distinct ecosystem. Low ecosystem divers

Diversity Explorer — one place, three lenses

06 — interactive Diversity Explorer — one place, three lenses Here's the same patch of Kestrel Ridge seen through each lens. Switch between the three levels and watch what "variety" means change. Drag to look around. Kestrel Ridge · field view loading… Pick a lens below. 3D unavailable. Genetic = one species, many slightly different individuals. Species = many different species together. Ecosystem = several distinct habitats side by side. Genetic Species Ecosystem

Richness vs evenness — proof in 3D

07 — the classic trap Richness vs evenness — proof in 3D Two heathland plots at Kestrel Ridge, each with the same five species (same richness). The only difference is how the 50 individuals are spread. Flip Plot B between balanced and dominated and watch the diversity verdict change — richness never moves. Two plots · 5 species each · 50 individuals each richness = 5 in both Each colour is one species. Bar height = how many individuals. 3D unavailable. Plot A: 10·10·10·10·10 (even). Plot B dominated: 46·1·1·1·1. Same richness (5), but Plot A has higher e

Classify it — which level?

◇ ACTIVE RECALL

08 — check yourself Classify it — which level? Drag each observation into the level it describes. (On a touch screen, tap a card then tap a bin.) Genetic diversity Species diversity Ecosystem diversity 0 / 7 placed

One connected web

09 — connect it One connected web The three levels aren't separate boxes — they're nested and interdependent . Genes make up species; species make up ecosystems. So a change at one level ripples to the others: Lose genetic diversity → the species can't adapt → it weakens or dies out. Lose a species → the ecosystem loses a role (a pollinator, a predator, a food source) → the ecosystem degrades. Lose an ecosystem → habitat vanishes for many species at once → species and genetic diversity both fall. That's why protecting a single threatened endemic species

Exam traps

⚠ EXAM TRAPS

10 — avoid these Exam traps ⚠ Trap 1 — mixing up genetic and species Genetic diversity is variation within one species ; species diversity is variation between species . "Within vs between" is the line examiners look for. ⚠ Trap 2 — stopping at richness Species diversity is richness AND evenness , not just the number of species. Many lost marks come from forgetting evenness. ⚠ Trap 3 — counting organisms for "ecosystem diversity" Ecosystem diversity = the variety of ecosystems / habitats , not the number of animals in one place. ⚠ Trap 4 — "biodiversity

Command words — do what's asked

◈ COMMAND WORDS

11 — read the verb Command words — do what's asked The verb in the question tells you how much to write. Marks are lost by describing when asked to distinguish , or listing when asked to explain . Define Give the precise meaning of the term. Short. Identify / State Name it — no explanation needed. Describe Give the features or characteristics. Distinguish Make the difference between two things explicit. Explain Say how or why — give reasons / cause and effect. Compare Give similarities and differences.

Structured answers · Point → Evidence → Link

12 — structure Structured answers · Point → Evidence → Link A clean short-answer structure: make your Point , support it with Evidence / an example , then Link back to the question. Here it is on a real prompt — reveal each line to study the moves. "Distinguish between genetic diversity and ecosystem diversity, with an example of each." (3 marks) Point Genetic diversity is the variety of alleles within one species, whereas ecosystem diversity is the variety of ecosystems across a region — they sit at different scales. Evidence For example, variation in d

Practice questions

★ PRACTICE & SELF-MARK

13 — exam practice Practice questions Original VCE-style questions. Attempt each one first, then reveal the weak answer, the model answer, and the marking guide — and self-mark to track your score. Question 1 define 2 marks Define the term biodiversity and identify its three categories. ✗ Weak answer ✓ Model answer ▣ Marking guide ✗ Weak answer "Biodiversity is the number of animals and plants in an area. The three types are big animals, small animals and plants." — defines it as a head-count, and the categories are wrong. ✓ Model answer Biodiversity is

Recap

✦ RECAP / CHEAT SHEET

14 — lock it in Recap The one table to memorise Category Variety of… Scope Kestrel Ridge example Genetic genes / alleles within one species disease-resistance genes varying among Helmeted Honeyeaters Species different species (richness + evenness) across species in an area 40 bird species recorded in the woodland survey Ecosystem ecosystems / habitats across a region creek wetland + grassy woodland + rocky ridge Biodiversity = the variety of all living things, at three levels. Genetic = within a species · Species = across species (richness and evenness)

Category	Variety of…	Scope	Kestrel Ridge example
Genetic	genes / alleles	within one species	disease-resistance genes varying among Helmeted Honeyeaters
Species	different species (richness + evenness)	across species in an area	40 bird species recorded in the woodland survey
Ecosystem	ecosystems / habitats	across a region	creek wetland + grassy woodland + rocky ridge

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT	→	E · EVIDENCE	→	L · LINK
answer the verb		example / mechanism		tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER	✓ STRONG / MODEL ANSWER
Biodiversity is the number of animals and plants in an area. The three types are big animals, small animals and plants." — defines it as a head-count, and the categories are wrong.	Biodiversity is the variety of all living things on Earth. Its three categories are genetic diversity (the variety of genes within a species), species diversity (the variety of species in an area) and ecosystem diversity (the variety of ecosystems in a region).
Genetic diversity is about genes and species diversity is about species." — no within/between contrast and no examples, so at most 1 of 3.	Genetic diversity is the variety of genes (alleles) within a single species or population — for example, variation in coat colour among a mob of eastern grey kangaroos. Species diversity is the variety of different species in an area — for example, 40 bird species recorded in one woodland survey. The key difference is scale: genetic diversity is variation inside one species, whereas species diversity is variation across many species.
They have the same diversity because both have 5 species." — confuses richness with diversity and ignores evenness entirely.	Both plots have the same species richness (5 species). However, species diversity depends on both richness and evenness. Plot A has high evenness — the 50 individuals are spread evenly (10 per species). Plot B is heavily dominated by one species (46 of 50 individuals), so its evenness is very low. Because Plot A combines the same richness with much higher evenness, Plot A has the greater species diversity .
It would be bad for the environment and kill the animals." — true but generic; it never addresses the three categories the question names.	Species diversity falls sharply — many native plant and animal species are replaced by essentially one tree species, lowering richness. Ecosystem diversity falls — varied native habitats (understorey, leaf litter, tree hollows) become one uniform plantation ecosystem. Genetic diversity is reduced — plantation pines grown from a few related stock hold far fewer alleles, and any surviving native species are left in small, less genetically varied populations.

✗ **WEAK ANSWER**

There aren't many left so it might die out, and all the diversity types are connected." — asserts the conclusion without explaining the genetic mechanism or how the levels actually link.

✓ **STRONG / MODEL ANSWER**

Low genetic diversity means the population carries few alleles , so it is the raw material for evolution that is missing — there are fewer individuals able to survive a new disease or environmental change, reducing the species' capacity to adapt . A small, isolated population also suffers inbreeding , which can lower fitness and survival. Together these raise the risk of extinction. The three categories are interdependent because they are nested : genes make up species, and species make up ecosystems. Healthy genetic diversity keeps a species viable; that species performs a role in its ecosystem (such as pollination or seed dispersal). So if genetic diversity is lost the species declines, an